CSE421 Assignment 01 [MSMA | Summer 2025]

Answer all the questions in this form. You can submit only once even if you submit by mistake. Make sure that your answers are put correctly by refreshing the page. Deadline: July 08, 2025 (Tuesday) 11:59:59pm

Q1. Which options of the following are correct?

1. Encapsulation is also done by the intermediary devices
2. Decapsulation is done only by the receiver end device
3. Protocols specify how to implement what needs to be done
4. Two layers can have modifications simultaneously

Answer: 1, 4

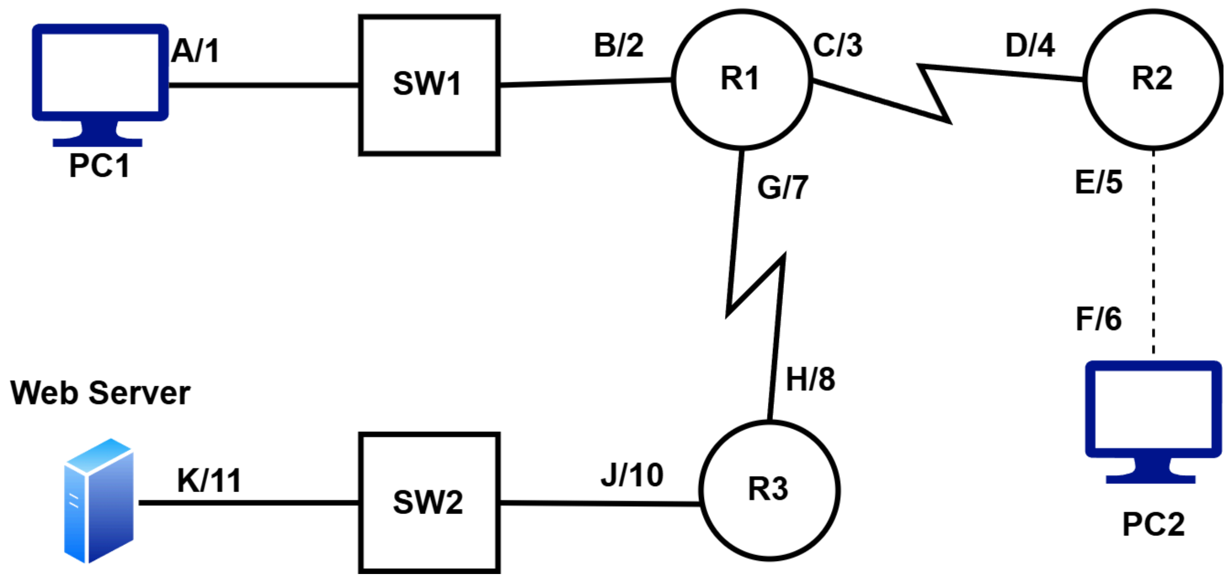
Q2. Transport layer is concerned with

1. Sending the segments of multiple processes to the network layer simultaneously
2. Sending a packet from one host to another host
3. Adding port address to identify the application
4. Adding sequence number to identify the order of the process

Answer: 1, 3

Answer Q3 to Q10 based on the following scenario.

A frame is leaving **R1** to reach **PC2**. The response packet was generated by the **web server**. Consider the alphabets to be physical addresses and the numbers to be logical addresses (given beside the figures).



Q3. Identify the source IP Address.

Answer: 11

Q4. Identify the destination IP Address.

Answer: 6

Q5. Identify the source MAC Address.

Answer: C

Q6. Identify the destination MAC Address.

Answer: D

Q7. How many times the IP address of the packet will change from source host to the destination host?

Answer: 0

Q8. How many times the MAC address of the packet will change from source host to the destination host?

Answer: 3

Q9. How many times the network layer will be traversed from source host to the destination host?

Answer: 5

Q10. How many times will the physical layer be traversed from source host to the destination host?

Answer: 6

Q11. Upon getting a request for a webpage by the browser, the server

1. sends all the objects at a time
2. sends the base file first
3. sends the html code of the base file
4. sends the html code for all the objects at a time

Answer: 3

Q12. Which method is used to transfer your searched keyword to the Google server.

1. GET
2. POST
3. PATCH
4. PUT

Answer: 1

Answer Q13 to Q17 based on the following scenario

You have requested to visit www.bracu.ac.bd website using a **persistent HTTP connection**. There are **X** objects in total. Among them, the first **14 objects are 6 MBytes each** and the remaining objects are **8 MBytes each**. Your computer's download speed is **140 Mbps**. It takes **43ms** for a small **TCP packe**t to go from your **computer to the BRACU server** and come back; **27ms** for a small **HTTP packet** to come from the **BRACU server to your Computer**. Your device waits **5 ms** after getting an **HTTP response** to send the **next TCP/HTTP request** (Consider this a **separate waiting delay**). The **total RTT** to fetch all the objects is **1717ms**.

Q13. Calculate the value of X.

Answer: 31

43 + 54X = 1717; X = 31

Q14. What is the total file transmission time? (in ms)

Answer: 12571.42857, 12571.429, 12571

(14 * 6 + 17 * 8) * 8 / 140 = 12.57142857 s = 12571.42857 ms

Q15. What is the total time taken to load the website? (in ms)

Answer: 14438.42875, 14438.429, 14438

1717 + 12571.42857 + 30 * 5 = 14438.42875

Q16. If the web browser uses non-persistent HTTP connection instead of persistent HTTP connection,

1. Total RTT would be higher
2. Total RTT would be lower
3. Object transmission time would be higher

4. Object transmission time would be lower
5. Total response time would be higher
6. Total response time would be lower

Answer: 1,5

The number of objects doesn't change, so the object transmission time will remain the same.

Q17. What is the total response time if the browser uses non-persistent HTTP connection? (in ms) [The number of objects will be same as found in Q10]

Answer: 15728.42875, 15728.429, 15728

(31 * 43 + 31 * 54) + 12571.42857 + 30 * 5 = 15728.42875